

Recession Compass — Methodology

v1.4 · 2026-05-02 · Internal / compliance distribution only

What

A single 0–100 number per region (US, Eurozone, UK, Global) that compresses 6–8 public economic indicators into one read. Higher = more recessionary signals firing. Bands map to plain-English copy:

Score	Phase
0–30	Skies are clear
30–60	Storm watch
60–80	Storm warning
80+	Storm in progress

Why

Retail investors get a flood of macro headlines and no scoreboard. The barometer answers one question — “*is a recession coming?*” — using only free public data, with every indicator and weight made explicit. No AI guesses. No black box. Same input → same number every time.

How — the calculation

Step 1. For each indicator, fetch the latest observation from FRED. Apply the configured transform: level (e.g., yield curve), year-over-year change in pp (e.g., unemployment), or YoY % (e.g., GDP).

Step 2. Map the raw value to a 0–100 sub-score via linear interpolation between two calibrated anchors:

`subScore = clamp(0, 100, (value - calm) / (alarm - calm) × 100)`

calm = value at which we'd score 0 (no concern). **alarm** = value at which we'd score 100 (recession-grade). Anchors come from each indicator's historical behaviour around past recessions.

Step 3. Composite is the weighted average of per-indicator sub-scores. Weights sum to 100 per region.

`score = Σ(weight_i × subScore_i) / Σ(weight_i)`

Step 4. A confidence band of $\pm 1.5 \times \text{stdev}(\text{sub-scores})$, capped at ± 35 , shades the needle. Wide band = indicators disagree; narrow band = they agree.

Calibration — indicator weights

Indicator	US	Eurozone	UK	Global
Yield curve (10y – 2y)	25	25 (DE)	25	—
Sahm rule	25	—	—	—
Unemployment YoY change	10	25	25	—
Initial jobless claims YoY	10	—	—	—
Industrial production YoY	10	—	—	—
OECD CLI (amplitude-adj.)	10	20	20	China 20, Japan 15
Real GDP YoY	5	12	12	—
Consumer sentiment / confidence	5	13	13	—
HICP YoY	—	5	—	—
BoE overnight rate (12m change)	—	—	5	—
US sub-score	—	—	—	35
Eurozone sub-score	—	—	—	30
Total	100	100	100	100

Sources (all FRED)

US: T10Y2Y, SAHMREALTIME, ICSA, INDPRO, USALOLITOAASTSAM, UNRATE, GDPC1, UMCSNT

Eurozone: IRLTLT01DEM156N – IR3TIB01DEM156N; mean(LRHUTTTT{DE,FR,IT,ES}M156S) for unemployment; mean(DEU,FRA,ITA LOLITOAASTSAM) for CLI; CLVMNACSCAB1GQEA19; CSCICP02EZM460S; CP0000EZ19M086NEST

UK: IRLTLT01GBM156N – IR3TIB01GBM156N; LRHUTTTTGBM156S; GBRLOLITOAASTSAM; NGDPRSAXDCGBQ; CSCICP02GBM460S; IUDSOIA

Global: composite(US) + composite(EU) + CHNLOLITOAASTSAM + JPNLOLITOAASTSAM

Recession bands shaded on the deep history chart: NBER (US), CEPR EABCDC (Eurozone), ONS (UK), OECD-dated global slowdowns (Global).

Staleness guard

Any indicator whose latest observation is older than **240 days** is flagged stale, given a neutral sub-score of 50, and surfaced with a 'Stale' badge in the gauge header. This prevents discontinued FRED mirrors (which is how the Philly Fed state-leading index, last published 2020-02, was silently corrupting the US score before v1.1).

What this score is not

Not a forecast model — anchors are hand-calibrated and transparent, not trained on history. Not a Bayesian probability — the 0–100 is a calibrated severity index. Not financial advice. Will sometimes peg high during slowdowns that don't get officially-dated as recessions, and vice versa.

Refresh cadence

GitHub Actions cron runs every 6 hours. Each run pulls the latest observations from FRED, recomputes all four region composites, commits resulting JSON to data/recession-{region}.json, which triggers a Vercel redeploy.

Trade-on-eToro baskets — region × phase mapping

The 'Trade on eToro' CTA in each region opens a basket sized to that region's current phase. Region = the *economy being hedged*, not the user's regulatory zone — e.g. UK Storm holds GBP-correlated assets (gilts, linkers, GBP cash, gold), even for a non-UK user, because the stress being hedged is UK-specific. US baskets stay in US-listed ETFs; EU/UK/Global baskets use UCITS-wrapped equivalents so they trade for European-domiciled users on eToro. Weights are % of the user-entered amount.

US — hedging US-economy risk (USD)

Phase	Holdings (% weight)
Clear	VTI 40, QQQ 25, VEA 15, VWO 10, SCHD 10
Watch	SCHD 30, VYM 20, VTI 20, BND 15, TIP 15
Warning	XLU 18, XLP 18, XLV 14, GLD 18, IEF 12, TIP 10, BND 10
Storm	BIL 28, TLT 22, TIP 15, SHV 10, GLD 20, XLU 5

Eurozone — hedging eurozone risk (EUR, UCITS)

Phase	Holdings (% weight)
Clear	EXSA 35, DBXD 20, EXSI 15, VEUR 20, EIMI 10
Watch	EUDV 28, EXSA 22, DBXD 15, EUNA 20, IBCI 15
Warning	EXH9 18, EXH4 14, EXH3 13, SGLN 18, EUNH 15, IBCI 12, EUNA 10
Storm	ERNE 25, EUNH 15, DBXG 15, IBCI 15, SGLN 20, EXH4 10

UK — hedging UK risk (GBP, UCITS)

Phase	Holdings (% weight)
Clear	VUKE 25, VMID 25, VUSA 20, VEUR 15, EIMI 15
Watch	VUKE 30, VHYG 20, VMID 15, IGLT 20, INXG 15
Warning	VUKE 18, EXH9 12, EXH4 10, SGLN 18, IGLT 18, INXG 14, ERNS 10
Storm	ERNS 28, IGLT 22, INXG 17, SGLN 23, EXH4 10

Global — hedging a global slowdown (USD/UCITS safe-haven complex)

Phase	Holdings (% weight)
Clear	VWRA 50, EQQQ 20, EIMI 15, VEUR 15
Watch	VHYL 30, VWRA 25, AGGH 20, VEUR 15, SGLN 10
Warning	EXH9 18, EXH4 12, SGLN 20, AGGH 15, IDTL 20, EIMB 15
Storm	IB01 30, IDTL 25, SGLN 25, AGGH 10, EXH4 10

Ticker dictionary

US-listed (US baskets only): VTI Vanguard Total Stock Market · QQQ Invesco Nasdaq-100 · VEA Vanguard FTSE Developed Mkts · VWO Vanguard FTSE EM · SCHD Schwab US Dividend Equity · VYM Vanguard High Dividend Yield · BND Vanguard Total Bond Market · TIP iShares TIPS · IEF iShares 7-10y Treasury · TLT iShares 20+y Treasury · BIL SPDR 1-3M T-Bill · SHV iShares Short Treasury · GLD SPDR Gold Shares · XLU/XLP/XLV Utilities/Staples/Healthcare Select Sector SPDRs.

UCITS (EU / UK / Global baskets): EXSA STOXX Europe 600 · EXSI STOXX Europe Mid 200 · EXH9/EXH4/EXH3 STOXX Europe 600 Health Care/Utilities/Personal & Household Goods · DBXD Xtrackers DAX · DBXG Xtrackers Eurozone Govt Bond 25+ · VEUR Vanguard FTSE Developed Europe · VUKE Vanguard FTSE 100 · VMID Vanguard FTSE 250 · VUSA Vanguard S&P 500 · VWRA Vanguard FTSE All-World (USD Acc) · EQQQ Invesco Nasdaq-100 UCITS · EIMI iShares Core MSCI EM IMI · EUDV SPDR S&P Euro Dividend Aristocrats · VHYL/VHYG Vanguard FTSE All-World High Dividend Yield · EUNA iShares Core € Corp Bond · EUNH iShares Euro Govt Bond 5.5-10.5y · IBCI iShares € Inflation Linked Govt Bond · ERNE iShares € Ultrashort · ERNS iShares £ Ultrashort · IGLT iShares Core UK Gilts · INXG iShares £ Index-Linked Gilts · IDTL iShares \$ Treasury 20+y · IB01 iShares \$ Treasury 0-1y · AGGH iShares Core Global Aggregate Bond (Hedged) · EIMB iShares JPM \$ EM Bond · SGLN iShares Physical Gold ETC.

Pre-launch verification

scripts/verify-baskets.ts (run via 'npm run verify:baskets', requires ETORO_API_KEY + ETORO_USER_KEY) hits eToro's /market-data/search for every ticker and reports ✓/✗ with the resolved instrumentId. Required green-light before any change to lib/baskets.ts ships to main.

Changelog

v1.4 (2026-05-02) — Region-aware ETF baskets. Region now means the economy being hedged (not user's regulatory zone). EU baskets hold euro-correlated assets (Bunds, € inflation-linked, € money-market). UK baskets hold UK-correlated assets (gilts, INXG linkers, £ money-market). US stays USD-listed. Global uses USD/UCITS world safe-haven complex. Added scripts/verify-baskets.ts as a pre-launch

ticker-availability guard.

v1.3 (2026-05-01) — Recalibrated anchors that didn't match historical recession behaviour. OECD CLIs (all 5 instances): calm 100.5 → 101, alarm 98/98.5 → 95 (matching GFC trough of 92–94). Sahm rule: alarm 0.8 → 0.5 (the textbook trigger). US industrial production: calm 4 → 3, alarm −4 → −10.

v1.2 (2026-05-01) — Removed EU and UK industrial production indicators (no live FRED mirror). Reweighted both regions across the remaining six indicators. Bumped staleness threshold from 180 to 240 days. Synthesised EU unemployment from DE/FR/IT/ES (the EZ aggregate stopped publishing in 2023). Swapped UK GDP from CLVMNACSCAB1GQUK (last 2020) to NGDPRSAXDCGBQ (current). Swapped UK Bank Rate from INTGSTGBM193N (last 2016) to IUDSOIA (SONIA).

v1.1 (2026-05-01) — Replaced 5 stale OECD CLI distributions across US/EU/UK/CN/JP with the maintained amplitude-adjusted variants. Eurozone CLI synthesised from DE/FR/IT (EA19 series gone). Recalibrated EU and UK consumer-confidence anchors from a 100-centred to a 0-centred scale. Added the staleness guard.

v1.0 (2026-05-01) — Initial release. US (FRED), Europe (FRED-mirrored ECB/Eurostat/OECD), UK (FRED-mirrored ONS/BoE/OECD), Global composite (US 35 / EU 30 / CN 20 / JP 15).